

PRACTICAL 2: MULTICLASS CLASSIFICATION USING DEEP NEURL NETWORK

```
import tensorflow as tf
from tensorflow import keras
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout
from tensorflow.keras.optimizers import RMSprop
from tensorflow.keras.datasets import mnist
import matplotlib.pyplot as plt
from sklearn import metrics

(x_train,y_train),(x_test,y_test)= mnist.load_data()

plt.imshow(x_train[0],cmap='gray')
plt.show()

print(x_train[0])

print('X_train shape',x_train.shape)
print('Y_train shape',y_train.shape)

print('X_test shape',x_test.shape)
print('Y_test shape',y_test.shape)

x_train= x_train.reshape(60000,784)
x_test= x_test.reshape(10000,784)
x_train=x_train.astype('float32')
x_test= x_test.astype('float32')
x_train/=255

num_classes=10
y_train= np.eye(num_classes)[y_train]  y_test= np.eye(num_classes)[y_test]
```

```
model = Sequential()
model.add(Dense(512, activation='relu', input_shape=(784,)))
model.add(Dropout(0.2))
model.add(Dense(512, activation='relu'))
model.add(Dropout(0.2))
model.add(Dense(num_classes, activation='softmax'))

model.compile(loss='categorical_crossentropy', optimizer='RMSprop', metrics=['accuracy'])

batch_size = 128
epochs = 20
history = model.fit(x_train, y_train, batch_size=batch_size,
                    epochs=epochs, verbose=1,
                    validation_data=(x_test, y_test))
score = model.evaluate(x_test, y_test, verbose=0)
print("Test loss:", score[0])
print("Test accuracy:", score[1])
```